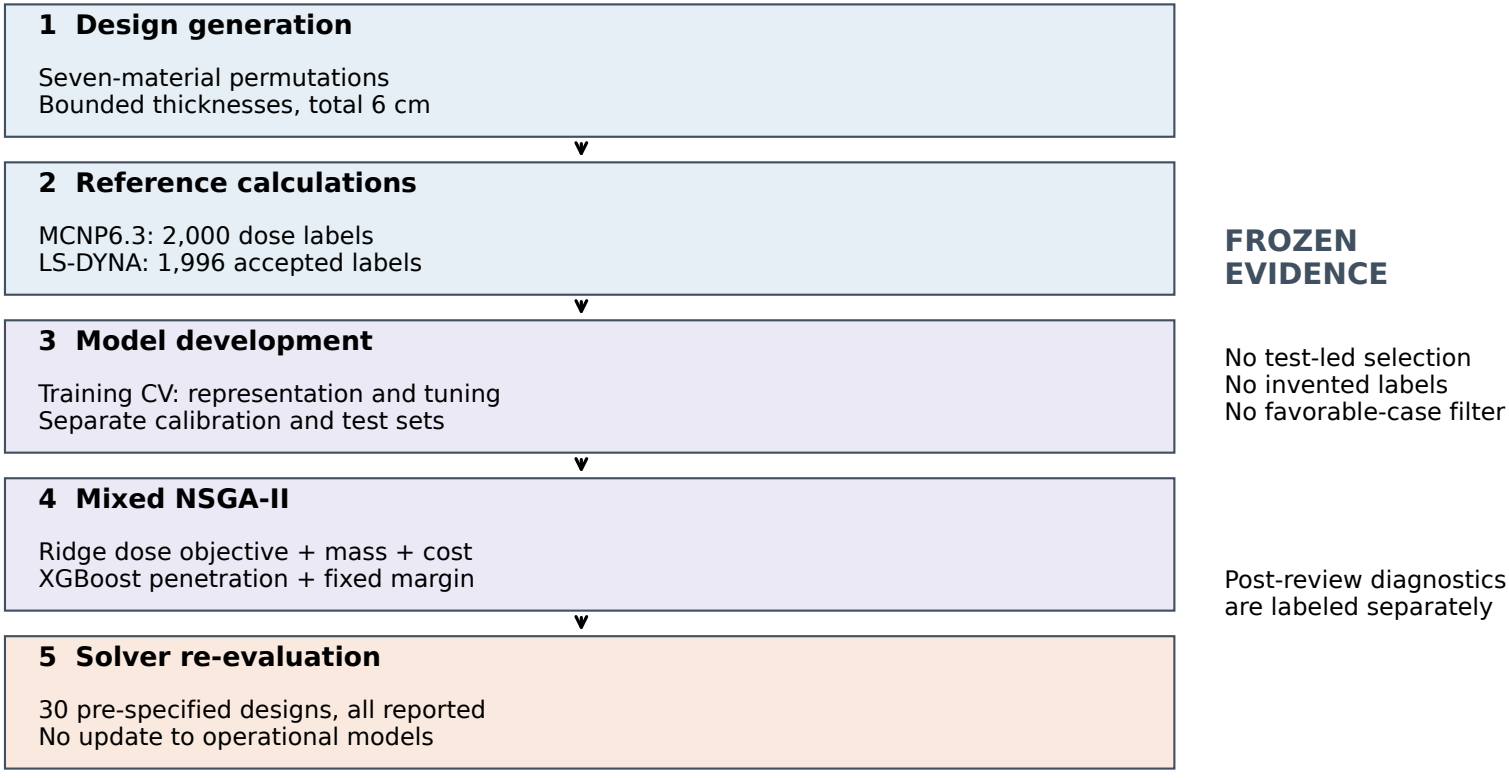
1 Design generation
Seven-material permutations
Bounded thicknesses, total 6 cm
2 Reference calculations
MCNP6.3: 2,000 dose labels
LS-DYNA: 1,996 accepted labels
3 Model development
Training CV: representation and tuning
Separate calibration and test sets
4 Mixed NSGA-II
Ridge dose objective + mass + cost
XGBoost penetration + fixed margin
5 Solver re-evaluation
30 pre-specified designs, all reported
No update to operational models
FROZEN EVIDENCE
No test-led selection
No invented labels
No favorable-case filter
Post-review diagnostics are labeled separately